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Effects of bulk impurity concentration on the reactivity of metal surface: Sticking of hydrogen molecules and atoms to polycrystalline Nb containing oxygen

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Nonmetallic impurities segregated onto metal surfaces are able to drastically decrease the chemical reactivity of metals. In the present paper, effects of *bulk* impurities on the reactivity of metallic surfaces were investigated in a wide temperature range on an example of the sticking of hydrogen molecules and atoms to Nb [polycrystalline, with mainly (100)] containing solute oxygen. At all the investigated surface temperatures, T_S (300–1400 K), we found the bulk oxygen concentration C_O to have a strong effect on the integral probability, α_{H_2}, of dissociative sticking of H_2 molecules followed by hydrogen solution in the metal lattice: α_{H_2} monotonically decreased by orders of magnitude with increasing C_O from 0.03 to 1.5 at. %. The sticking coefficient α_{H_2} was found to depend on T_S but not on the gas temperature. The effect of C_O on α_{H_2} is explained by the presence of oxygen-free sites (holes in coverage) serving as active centers of the surface reaction in the oxygen monolayer upon Nb. In contrast to H_2 molecules, H atoms were found to stick to, and be dissolved in, oxygen-covered Nb with a probability comparable to 1, depending neither on C_O nor on T_S. This proves that, unlike H_2 molecules, H atoms do stick to be dissolved mainly through regular surface sites covered by oxygen and not through the holes in coverage. © *2007 American Institute of Physics*. [DOI: 10.1063/1.2804874]

I. INTRODUCTION

Reactivity toward gaseous molecules is important for the application of metallic materials as catalysts, hydrogen permeation membranes and storage materials, nonevaporable getters, and so on. Metal surfaces in their "natural" state, however, are typically covered by thin oxide films that severely degrade the reactivity. Hence, metallic materials for those purposes are usually "activated" by heating in an inert atmosphere, vacuum, or reducing atmosphere such as H_2. The oxide films can be removed by such treatments through the dissolution of oxygen in the bulk or through reduction, and the reactivity can be improved for certain extent. Nevertheless, clean surfaces are still not obtained in most cases. Namely, the surfaces remain to be covered by an adlayer of oxygen or other nonmetallic impurities segregated from the bulk, and the reactivity of surfaces is still much lower than that of clean surfaces. For example, an oxygen monolayer present on the surface of group V metals causes an orders-of-magnitude decrease in the probability of dissociative sticking of such diatomic molecules as H_2,[1–6] N_2,[7] and O_2,[8] as well as in the probability of decomposition of hydrocarbons C_nH_m.[9]

If the metal temperature T_S is sufficiently high, the equilibrium is attained between the impurity on the surface and in the bulk. According to the Langmuir-McLean equation, the correlation between the surface impurity coverage, θ_O, and the impurity concentration in the bulk, C_O, is described as

$$\frac{\theta_O}{1-\theta_O} = k_O C_O \exp\left(-\frac{\Delta H_{seg}}{RT_S}\right), \tag{1}$$

where ΔH_{seg} is the heat of surface segregation of the impurity and k_O is the constant coefficient. Usually, ΔH_{seg} takes a large negative value, and hence θ_O is comparable to unity even at very low C_O (e.g., at C_O in hundredths of at. % or even smaller). Nevertheless, a certain portion of the surface, $1-\theta_O$, still remains free from the impurity at any T_S and C_O. In other words, even a *monolayer* coverage ($\theta_O \approx 1$) contains a definite number of stochastically distributed holes (the surface sites free from the impurity).

One can consider two different concepts with regard to the reactivity of a metal surface covered by nonmetallic im-

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purities segregated from the bulk. The difference will be in the sort of surface sites supposed to serve as the active centers of surface reactions.

(1) The "holes in the coverage" concept: the surface sites free of impurity (holes in coverage) form active centers, e.g., double centers with a $(1-\theta_O)^2$ concentration for reactions involving diatomic molecules. This mechanism appears to be dominant at higher T_S where, according to Eq. (1), a considerable portion of clean surface opens up. Since $1-\theta_O$ depends on C_O, bulk impurity concentration will govern reaction kinetics in this case, as demonstrated by Hörz and Fromm,[9] who reported the radical influence of bulk concentrations of oxygen and nitrogen on the reactivity of polycrystalline Nb and Ta toward the decomposition of C_nH_m molecules at $T_S > 1200$ °C. According to Eq. (1), however, the fraction of coverage holes, $1-\theta_O$, is negligible at lower T_S. For example, $(1-\theta_O)^2$ should be in a range of 10^{-10} at $T_S \sim 500$ K in the O–Nb system even at C_O as low as 0.1 at. % (with $|\Delta H_{seg}| = 71$kJ/mol).[10] Nevertheless, H_2 molecules of room temperature were found to stick to Nb surface covered by segregated oxygen with a probability as high as 4×10^{-5} at $T_S = 500$ K.[11]

(2) The "holes in the barrier" concept: here, holes are regarded to be inherent to the regular surface sites covered by impurity,[4] i.e., the holes are not the "stochastic" sites with the lack of impurity atoms but such area in *each* surface unit cell where the barrier is lower than in other parts for an incident molecule with particular orientation and internal state. For example, Eibl and Winkler[4,12] who measured the sticking coefficient of D_2 molecules on V(111) covered by oxygen and sulfur at $T_S = 77$ K and T_g = room temperature attributed a rather large value of sticking coefficient, 10^{-4}, to such holes in the barrier (T_g indicates the gas temperature). The presence of holes in the barrier means that the potential energy surface (PES) is highly corrugated.[4] With a more general approach proposed for the H_2–Cu(111) system,[13] the holes in the barrier are not static: areas with a lower barrier can emerge as a result of vibration of surface atoms, and therefore the concentration of holes in the barrier should depend on T_S,[13] and correspondingly so should the reaction probability.

Which of the two mechanisms defines the reactivity of surface covered by a nonmetallic impurity, e.g., for the dissociative sticking of H_2 molecules, at a *relatively low* T_S where one should expect a virtually saturated coverage ($\theta_O \approx 1$)? One can find the answer by the presence or absence of a dependence of sticking coefficient, α_{H_2}, on the bulk impurity concentration. Namely, C_O should have no effect, if H_2 sticking occurs through a regular surface site covered by impurity, i.e., via *holes in barrier*. If, vice versa, α_{H_2} is governed by C_O, then the *holes in coverage* are responsible for sticking.

The answer to the question is important for the formulation of the theoretical vision of the reactions of simplest molecules (H_2) with a "real" surface (covered by nonmetallic impurities).[14,15] For instance, if H_2 sticking occurs mainly through the holes in coverage even at relatively low T_S, it may mean that no *holes in the barrier* actually exist, i.e., that PES is not so corrugated for H_2 molecule sticking. In this case, typically observed near-cosine angle distribution of the desorbed H_2 molecules[4] can also be explained by the holes in coverage.

If the concentration of holes in a near-monolayer coverage ($\theta_O \approx 1$) actually determines the surface reactivity, and if it is defined by C_O, the surface reactivity can be controlled by changing C_O; for example, the reaction kinetics of hydrogen storage materials can be improved by removing impurities from the bulk and, vice versa, the suppression of the recombinative desorption of hydrogen by doping nonmetallic impurities can enhance the performance of superpermeable membranes.[1–3] Besides, the measurement of C_O can be a method to predict the surface reactivity in addition to the conventional methods of surface analysis such as x-ray photoelectron spectroscopy and Auger electron spectroscopy (AES) incapable to measure a small concentration of holes in a near-monolayer coverage.

The evidence of the important role played by bulk impurity concentration in the H_2 molecule sticking/desorption at relatively low T_S was observed in the previous study.[16] The rate constant k_r of recombinative desorption of D_2 molecules from polycrystalline Nb demonstrated the dependence on C_O over a wide range at all the investigated temperatures, including T_S as low as 620 K.[16]

Considering the importance of the effect of bulk impurities on the reactivity of metal surfaces, we undertook a systematic study of the sticking of H_2 molecules and H atoms to the polycrystalline Nb [mainly (100)] containing oxygen as bulk impurity over possibly wide ranges of oxygen concentrations and temperatures. The mechanisms defining the reactivity of oxygen-covered Nb surface toward the sticking of H_2 molecules and H atoms were discussed.

II. EXPERIMENTAL

A. Apparatus

The experiments were performed in an ultrahigh vacuum (UHV) apparatus schematically shown in Fig. 1. Two types of samples were employed: membrane and ribbon. A *membrane sample* of tubular shape hermetically separates the outside and inside vacuum chambers; both chambers are continuously pumped by turbo-molecular and sputter-ion pumps and are equipped with ionization gauges and quadrupole mass analyzers calibrated by a diaphragm gauge. *A ribbon sample* is located in an auxiliary chamber connected to the outside chamber with a short duct of high conductivity. Temperatures of both samples are regulated by Ohmic heating from room temperature to the melting point. Hydrogen gas (as well as O_2) is introduced into the outside/auxiliary or inside chamber through variable-leak valves.

Atomic hydrogen is produced by dissociation of H_2 molecules at a set of nine incandescent filaments made of 0.8 mm Ta wire and installed at the top and bottom of the outside chamber over its perimeter. Molybdenum shields prevent the deposition of filament material onto the sample

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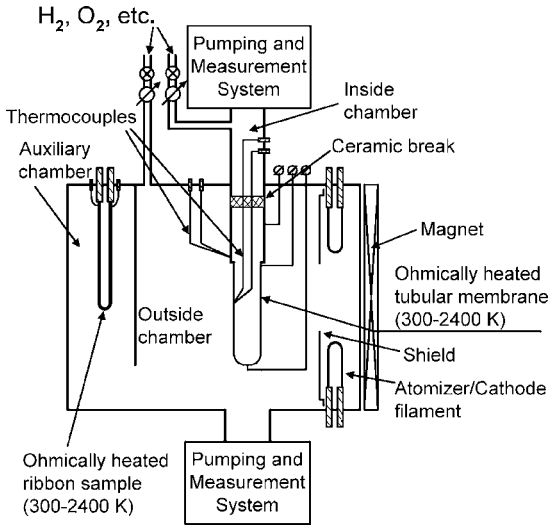


FIG. 1. Schematic of experimental apparatus.

and heating of sample by radiation. *Hydrogen plasma* filling the outside chamber is generated for sputter cleaning of the membrane sample by dc glow discharge between the Ta filaments serving as cathodes and the chamber wall.

The apparatus can be baked at 520 K. The pressure of residual gases after baking is $\sim 10^{-8}$ Pa (mainly H_2, then CO). The outer wall of the chamber is cooled by water when the sample(s) and filaments are heated to sustain sufficiently low pressures of residual gases.

B. Sample design and treatment

1. Tubular membrane

A tubular membrane (10 mm diameter, 100 mm length, and 0.1 mm thickness) made of 99.95% purity polycrystalline Nb was used as a sample. Temperature of this sample was measured with a W-Re (5%)/W-Re (20%) thermocouple located inside the tubular membrane and with an optical pyrometer. One end of the tubular membrane was dead ended by welding, while the other one was welded to a stainless steel tube transition. The membrane was electrically isolated from the chamber with a ceramic break. Special measures were taken to secure the temperature uniformity over the membrane sample length. First, the piece of stainless steel transition immediately contacting the membrane was resistively heated, independently of the sample (Fig. 1), and its temperature was equalized with that of the membrane by using separate thermocouple. Second, the electric leads connected to the membrane sample ends were made in the form of long Nb ribbons Ohmically heated to the same temperature as the membrane.

The sample was heated in UHV to temperatures exceeding 1770 K prior to the start of the experiments. The first aim of the heat treatment was to ensure that no carbide layers are present on the surfaces. Carbide layers on Nb are known to suppress the exchange of oxygen between the surface and bulk if such layers were present[17,18] and to be removed by heating in vacuum through the release of CO.[19] Secondly, this heat treatment resulted in the recrystallization and grain growth; crystal grains of 0.1–1 mm size appearing along the sample plane and spreading over the whole 0.1 mm sample thickness were obtained. The analyses by means of x-ray diffraction and electron backscatter diffraction showed that the surfaces mainly consisted of Nb(100) (~90%) inclined at a 0.8°–1.9° angle to the surfaces and also Nb(211) (~10%) inclined at ~2.2°.

Oxygen concentration in the sample was controlled in the following manner. First, the concentration of oxygen initially present in the sample was reduced by ion sputtering in hydrogen plasma with applying a negative bias (-600 V, 0.2 mA/cm^2) to the sample kept at an elevated temperature (1110 K) to secure a continuous surface segregation of oxygen in the course of sputtering. The degree of the purging of oxygen was checked from time to time by measuring the sticking coefficient α_{H_2} (see below). It has increased several orders of magnitude during the first ~5 h of sputtering, but the increase stopped at $\alpha_{H_2} \approx 10^{-3}-10^{-2}$, i.e., rather far from the magnitude in a few tenths as typical of a clean surface (e.g., $\alpha_{H_2} \approx 0.25$ was obtained in the present work, see Sec. III A). The residual concentration of solute oxygen was estimated to be ~0.03 at. % by extrapolation to lower concentrations of the observed dependence of α_{H_2} on C_O (see Sec. III A). Then, oxygen was introduced into the sample *in situ* by gas absorption method.[16] Namely, a steady-state O_2 pressure (in the range of $10^{-6}-10^{-2}$ Pa) was first established at continuous O_2 flow and pumping, at which the sample was kept at room temperature. No more than a few monolayers of oxygen could be absorbed by Nb under such conditions.[8] Heating of the sample to 1270–1370 K resulted in oxygen uptake, which was indicated by the reduction in O_2 pressure. The amount of absorbed oxygen could be routinely controlled and calculated from the extent of pressure reduction and exposure time. The sample was heated up to ~1770 K after the absorption of each single oxygen dose to ensure a uniform distribution of solute oxygen over the sample thickness.

2. Ribbon sample

The ribbon sample (7.2 mm width, 370 mm length, and 0.1 mm thickness) made of the same Nb foil as the membrane sample was used for the measurement of α_{H_2} on a clean surface. Deep cleaning of the samples from solute oxygen was performed by cyclic heating at 2370 K in UHV.[7,10] The rate of hydrogen absorption was examined after each heating cycle. A drastic increase was observed after every cycle at the beginning, but the rise has stopped upon the sticking coefficient reaching its value characteristic of a clean surface, $\alpha_{H_2} \approx 0.25$. This routine procedure results in the Nb specimen surface remaining free from segregated oxygen even at elevated temperatures.

C. Measurements of the sticking coefficients of H_2 molecules and H atoms

We used several different methods to take measurements of the sticking coefficients over wide ranges of C_O and gas/sample temperature.

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1. Permeation method to study the sticking of H_2 molecules

It should be noted in the first place that the permeation methods employed in this work are fundamentally different from those often used in other studies in this field. Usually, permeation of hydrogen is merely employed to produce a stationary flux desorbed from the sample surface, and the main purpose is an analysis of state/energy/angle distributions of the desorbing particles (e.g., Ref. 20). Sticking probabilities can be found from such desorption experiments by using the detailed balance principle (e.g., Ref. 13). In the present work, the integral sticking coefficient was *directly* determined from permeation because the dissociative sticking is one of the sequential steps of the permeation process. If the sticking is the rate-limiting step in permeation, and if the two membrane surfaces are identical, the permeation flux in steady state, $J^{\infty}_{\mathrm{perm}}$, equals to a half of the "sticking flux" $J_{\mathrm{inc}}\alpha_{\mathrm{H_2}}$,[1,2] and hence

$$\alpha_{\mathrm{H_2}} = 2J^{\infty}_{\mathrm{perm}}/J_{\mathrm{inc}}. \tag{2}$$

The ranges of temperature, flux, and sample thickness where Eq. (2) is valid become wider with decreasing $\alpha_{\mathrm{H_2}}$.[1] The group V metals are particularly suitable for measuring $\alpha_{\mathrm{H_2}}$ by this method due to several reasons including high diffusivity and solubility of hydrogen.[1,2,19] For instance, in the case of a Nb membrane of 0.1 mm thickness, Eq. (2) is true even for a clean surface below ~1300 K (and even at higher T_S, if the surface is covered by impurities).[1,2]

The permeation method allows direct measurements of $\alpha_{\mathrm{H_2}}$ over a wide temperature range, including high temperatures (e.g., up to 1620 K as described below). Another advantage of the method is its extremely high sensitivity due to a low background pressure at the downstream side of the membrane sample. Therefore, the method is especially effective at measuring small $\alpha_{\mathrm{H_2}}$, i.e., at investigating the surfaces covered by nonmetallic impurities. These unique capabilities of the permeation method allowed to obtain data on $\alpha_{\mathrm{H_2}}$ over wide ranges of T_S and C_{O} as presented below.

The permeation experiment with tubular membrane allows to distinguish *the roles of the surface* (T_S) *and gas* (T_g) temperatures in the dissociative sticking of H_2 molecules. If the gas is introduced at the outside of the tubular membrane (Fig. 1), $T_g \cong$ ambient temperature. Thus, with varying the sample temperature, one can find the dependence of $\alpha_{\mathrm{H_2}}$ only on T_S. If, however, the gas is introduced into the inner membrane hollow (Fig. 1), one will find $\alpha_{\mathrm{H_2}}(T)$ under thermostatic conditions ($T_g \cong T_S \equiv T$). Thus, relative effects of T_g and T_S on $\alpha_{\mathrm{H_2}}$ can be found by comparing the permeability in the two opposite directions. Another way to find $\alpha_{\mathrm{H_2}}(T)$ corresponding to thermostatic conditions is to determine $\alpha_{\mathrm{H_2}}$ from the rate constant of recombinative desorption k_r with using the detailed balance principle,[1]

$$\alpha_{\mathrm{H_2}} = \frac{z}{S^2}k_r, \tag{3}$$

where z is the gas kinetic theory coefficient and S is the solubility constant of hydrogen in Nb.[19] In the permeation regime in question, k_r can be found from the lag time τ of the establishment of permeation flux through a membrane having identical up- and downstream surfaces (Fig. 2),[1,21]

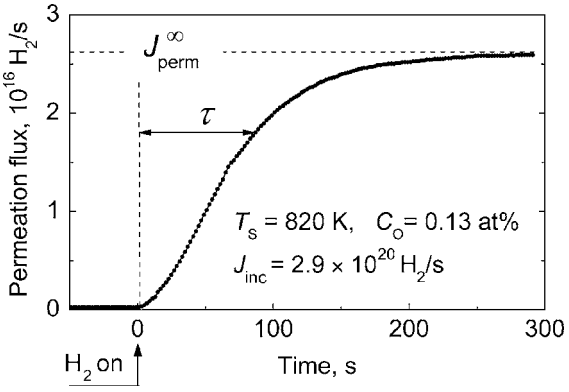


FIG. 2. Typical example of the establishment of steady-state permeation through a 0.1 mm thick Nb membrane.

$$J_{\mathrm{perm}}(t) = J^{\infty}_{\mathrm{perm}} \tanh^2(t/\tau), \quad \tau = \frac{L}{4\sqrt{k_r J^{\infty}_{\mathrm{perm}}}}, \tag{4}$$

where t is time and L is the membrane thickness. The values of $\alpha_{\mathrm{H_2}}(T)$ at $T_g = T_S$ found from τ by Eqs. (3) and (4) can be compared with $\alpha_{\mathrm{H_2}}(T_S)$ at $T_g = 300$ K.

The lag time measurement can also provide one more evidence that sticking (but not diffusion) is actually the rate-limiting step in permeation. If the permeation is limited by the bulk diffusion process, the lag time is expressed as $\tau = L^2/6D_{\mathrm{H}}$, where D_{H} is the hydrogen diffusion coefficient. The lag time actually observed in all our permeation experiments greatly exceeded $L^2/6D_{\mathrm{H}}$, indicating the surface-limited permeation, for which Eqs. (2) and (3) are valid. For example, the value of $L^2/6D_{\mathrm{H}}$ at 820 K evaluated with the data on D_{H} (Ref. 19) was 0.15 s, while the observed τ was 86 s (Fig. 2).

This permeation method was applied to examine the correlation between $\alpha_{\mathrm{H_2}}$ and C_{O} in the range of T_S from 470–670 K (depending on C_{O}) to 1620 K. The upstream pressure of H_2 was adjusted from 0.1 Pa (at lowest C_{O}) to 10 Pa (at highest C_{O}). Although we could not measure $\alpha_{\mathrm{H_2}}$ at the initial stage of sample interaction with hydrogen by the permeation techniques, the obtained values of $\alpha_{\mathrm{H_2}}$ should not noticeably differ from the initial magnitude because both the metal bulk and surface were far from saturation by hydrogen under the present experimental conditions (see Sec. II C 3).

Identity of the states of two surfaces was checked by thermal desorption experiments. The membrane sample was exposed to H_2 at a certain T_S, and then thermal desorption experiments were carried out by raising T_S. Desorption fluxes from the outside and inside surfaces were always close to the other.

2. Permeation method to study the sticking of H atoms

Heating the filaments (Fig. 1) up to ~1770 K (at H_2 pressure 10^{-6}–10^{-4} Pa) results in the generation of atomic hydrogen and consequently in the radical increase in perme-

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ation flux.[1,21,22] One can evaluate the sticking coefficient for atoms, α_H, in the same way as α_{H_2} [see Eq. (2)],

$$\alpha_H = 4(J^{\infty}_{perm})_H/(J_{inc})_H, \quad (5)$$

where $(J^{\infty}_{perm})_H$ is the atom-driven permeation flux (counted in molecules) and $(J_{inc})_H$ is an incident flux of atoms (counted in atoms). We can estimate only roughly the $(J_{inc})_H$ magnitude in this work, but $(J_{inc})_H$ remains constant at changing C_O and T_S. Thus, the dependence of α_H on C_O and T_S can be found from $(J^{\infty}_{perm})_H$, at least with the accuracy of an uncertain constant coefficient. More exact absolute α_H values can be found in atomic beam experiments.[22,23]

3. Absorption method for measurements of α_{H_2} at low T_S

The lag time of the establishment of permeation (Fig. 2) became too long at temperatures below 470–670 K (depending on C_O). Hence, *absorption* method was used in such temperature range. First, the membrane sample was exposed to H_2, at a definite low T_S, with hydrogen uptake well below the equilibrium concentration and also below the level of hydride formation. Then, the absorbed hydrogen was desorbed at sample heating to determine α_{H_2} as the ratio of the amounts of desorbed and incident (at loading) H_2 molecules.

The amount of absorbed hydrogen was proportional to both exposure time and H_2 pressure. Therefore, we can assume that the surface of the membrane sample precovered by oxygen was never saturated by hydrogen under the present conditions. It means that the sticking coefficient found by this method does not noticeably differ from that at zero hydrogen coverage (the "initial sticking coefficient"). Note that it is all the more true for the permeation experiments carried out at higher T_S. Additional arguments in favor of that the surface is not saturated by hydrogen will be presented below (Sec. IV A 3).

4. Measurements of α_{H_2} for clean surface

In the case of clean surface, molecule-driven permeation is too large to be measured because of a limited geometrical conductance of the present tubular membrane. Hence, the ribbon sample was used in the measurements for clean surface. Due to a very large α_{H_2}, the outgassed ribbon sample absorbed H_2 with an initial volumetric speed as high as in a few hundreds of dm^3/s. The values of α_{H_2} were evaluated from this temporary extra pumping at $T_S < 990$ K and H_2 pressures of 10^{-6}–10^{-5} Pa in the manner described in Ref. 24.

In the range of higher temperatures, α_{H_2} was measured by the method proposed by Brennan,[25] based on the fact that almost the whole molecular flux sticking to the surface ($\alpha_{H_2} J_{inc}$) is escaping back into the gas as a flux of H atoms, $J_H (J_H = 2\alpha_{H_2} J_{inc})$, if the surface is hot enough and the pressure p is sufficiently low (e.g., if $p < 10^{-4}$ Pa at $T_S > 1770$ K). The atoms produced in this process are trapped by the chamber wall[25] and one can evaluate α_{H_2} by this extra pumping. In the present experiment, H atoms produced by heating the ribbon sample to 1790 K were effectively trapped by the surrounding walls (Fig. 1) covered by the Nb film evaporated during the sample cleaning. Volumetric speed of this extra pumping was a few hundreds of dm^3/s.

D. AES investigation of oxygen segregation onto the surface

Surface analysis by AES was performed in a separate UHV device. Typical residual gas pressure was 5×10^{-7} Pa.

Sheet-type samples at $C_O = 0.1$, 0.3, 1, and 2 at. % were prepared from the same Nb foil as the above-mentioned membrane and ribbon samples. These samples were first cleaned by heating in an UHV at 2370 K, and then oxygen was introduced up to the above-mentioned concentrations in the manner described in Sec. II B 1.

The samples were mounted on a molybdenum holder and heated by electron bombardment from the backside. The temperature was measured with an optical pyrometer calibrated by using a similar sample having a thermocouple fixed to it. The samples were first heated to 1270 K in the AES apparatus before the analysis in order to remove the oxide film formed during the sample transportation by dissolution of oxygen into the bulk and level off oxygen concentration over the whole specimen thickness. Then, the temperature of the sample was kept at a given value until the stable oxygen coverage has been attained. The kinetic energy of primary electron beam was adjusted to be 3 keV. The diameter of the beam was ~5 μm.

III. RESULTS

A. Influence of bulk oxygen concentration and sample temperature on sticking probability

Figure 3 shows the dependence of α_{H_2} on T_S for the clean surface of the Nb ribbon sample and that for the surface of oxygen-doped membrane sample. The data for the former were obtained in the manner described in Sec. II C 4 and those for the latter were acquired by the methods mentioned in Secs. II C 1 and II C 3. The data shown in this figure indicate the values obtained at T_g=room temperature. The presented sticking coefficients do not noticeably differ from that at zero hydrogen coverage (the initial sticking coefficient), as described in Secs. II C 1 and II C 3.

The values of α_{H_2} for the clean surface was close to 0.25 being independent of T_S in the range as wide as 300–1800 K. On the other hand, α_{H_2} for oxygen-doped sample was radically smaller than that for clean surface and dependent on T_S with a characteristic bend of the Arrhenius plot. It appears to be most important in the context of this work that α_{H_2} decreased in orders of magnitude with increasing C_O in a whole range of T_S examined.

The data on α_{H_2} for the oxygen-doped membrane sample at room temperature are probably less accurate [α_{H_2}(300 K) may be underestimated] because the sample was exposed to the residual gas for long time after the preparation at elevated temperatures. We are still presenting in this figure the data on α_{H_2} at room temperature, which exhibit a dependence on C_O similar to that at higher T_S.

The dependence of α_{H_2} on the thermal history of the sample was carefully examined to understand whether or not

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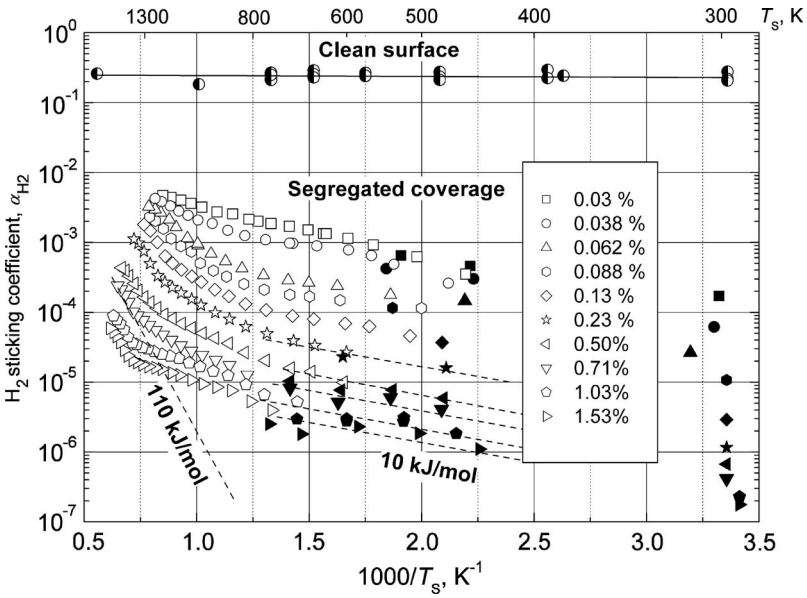


FIG. 3. Temperature dependence of α_{H_2} for the Nb with clean surface and with the surface covered by oxygen segregated from the bulk at different oxygen concentrations C_O (the values of C_O are indicated in the inset). Open and solid symbols correspond to the data obtained by permeation and absorption experiments, respectively. The data presented here were obtained at varying T_S with T_g=const=300 K.

the presented data pertain to a definite state of the Nb surface fully determined by T_S and C_O. That is definitely so in the range of high enough T_S and C_O where there is an equilibrium between the surface and bulk oxygen. However, the establishment of such equilibrium may be very slow at lower T_S and C_O, and it is practically never reached. In this range, oxygen coverage, and correspondingly α_{H_2}, may be different at one and the same T_S and C_O, depending on thermal history. Figure 4 shows temperature dependencies of α_{H_2} at several selected values of C_O for two different measurement modes. In the first case (the open symbols), the membrane sample was quickly cooled down from 1370 to ~470 K and then α_{H_2} was measured by permeation experiments with T_S elevated from ~470 to 990 K in a step-by-step manner. The permeation measurement of α_{H_2} at 990 K needed a few tens of seconds. When, however, the sample was kept at 990 K for a longer time, the α_{H_2} started to decrease and reached a new stable level in ~1 h. After keeping the sample at 990 K for 1 h, the temperature dependence of α_{H_2} was measured again, but with T_S descending down (the solid symbols in Fig. 4). At low C_O (0.062 at. % in this figure), the stable α_{H_2} level (the solid symbols at 600–700 K) turned out to be almost twice as low as the value before the equilibration at 990 K. The effect of heat treatment at 990 K became smaller with increasing C_O and completely disappeared at C_O =0.23 at. %. Taking into account these observations, the data on α_{H_2} at C_O<0.23 at. % presented in Fig. 3 were obtained after keeping the sample at 990 K for 1 h. Heat treatments at lower T_S, e.g., at 770 K, resulted in no noticeable change in α_{H_2} at any C_O. Thus, the presented $\alpha_{H_2}(T_S, C_O)$ must correspond to stable states of the surface of Nb well determined by T_S and C_O over the whole ranges of T_S and C_O investigated.

The data on α_{H_2} at selected temperatures are plotted against C_O in Fig. 5. In the high T_S region where the slope in Fig. 3 is large, α_{H_2} is inversely proportional to C_O^2. Weaker C_O dependence of α_{H_2} was observed at the lower temperature region.

In this figure, the data on the sticking coefficient of atomic hydrogen α_H are also plotted for comparison. First of

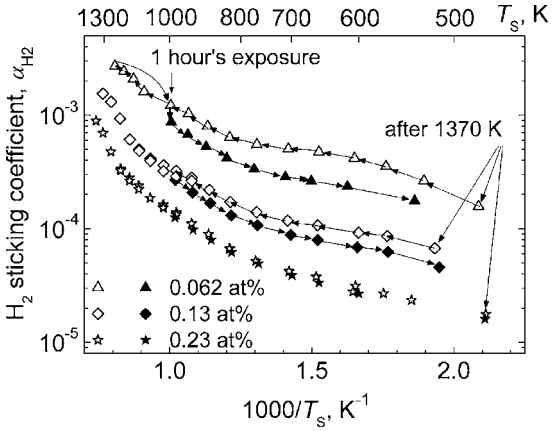


FIG. 4. The effects of sample heating at 990 K on temperature dependencies of α_{H_2} at several concentrations of solute oxygen. Open symbols refer to α_{H_2} (T_S) measured at temperature being ramped up after fast cooling of the sample from 1370 to 470 K, and solid symbols to α_{H_2} (T_S) at temperature ramp down after the sample heating at 990 K for 1 h.

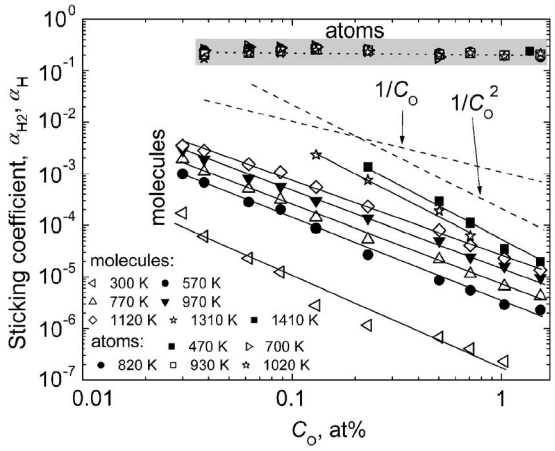


FIG. 5. Dependence of the sticking coefficients of H atoms (α_H) and H_2 molecules (α_{H_2}) on C_O at several temperatures.

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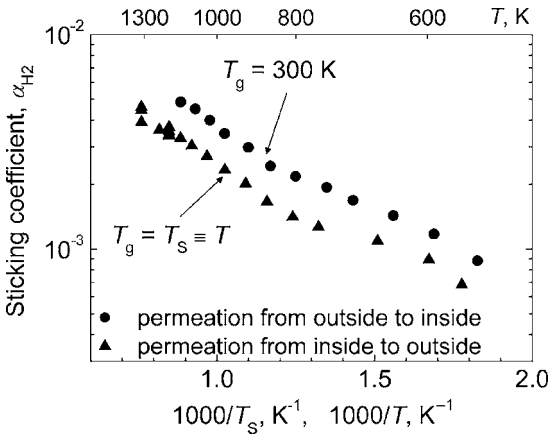


FIG. 6. Temperature dependence of α_{H_2} on Nb surface covered by segregated oxygen at varying only T_S (T_g=const=300 K) and under thermostatic conditions ($T_g \cong T_S \equiv T$). The concentrations of oxygen in the bulk C_O were the same (0.03 at. %) in both cases.

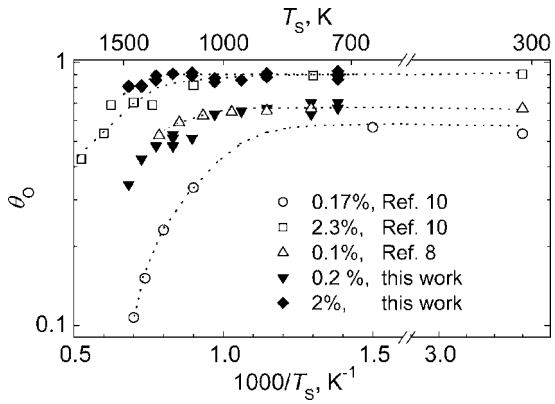


FIG. 7. Temperature dependence of the oxygen coverage θ_O on Nb due to surface segregation of oxygen from the bulk at different C_O (indicated in at. %).

all, α_H depends neither on T_S nor on C_O—in contrast to α_{H_2}. As to the absolute value of α_H, it was not directly measured (Sec. II C 2): only its lower limit could be estimated from the present experiments. Atom-driven permeation flux $[(J^{\infty}_{perm})_H$, see Eq. (5)] exceeded the largest permeation flux driven by molecules in the present experiments, even though the incident atomic flux $(J_{inc})_H$ was at least a few ten times smaller than the incident molecular flux (because the degree of H_2 dissociation was no greater than a few percent). Hence, α_H was at least a few ten times greater than the maximum α_{H_2}, i.e., than 5×10^{-3} (see Fig. 3), and therefore one can estimate the α_H magnitude to be in the order of at least 0.1. Atomic beam experiments with a number of metals covered by nonmetallic monolayers yielded α_H value in the range of 0.1–0.4 at $T_S \geqslant 298$ K,[1,4,5,22,26,27] while the data on the superpermeation of H atoms through Nb covered by oxygen monolayer[1–3,17,18] are most self-consistent at $\alpha_H \approx 0.2$–0.3. Taking that into account, we set $\alpha_H \approx 0.25$ in Fig. 5 and indicate by a gray band the uncertainty interval within which the $\alpha_H(T_S, C_O)$ plot may float, with keeping independence of C_O and T_S.

B. H_2 sticking under thermostatic conditions: Effects of gas and surface temperature

Temperature dependencies of α_{H_2} at $T_g \cong T_S \equiv T$ (thermostatic conditions) and at T_g=const=300 K (with only T_S varying) are presented in Fig. 6 for one value of C_O (0.03 at. %) to provide a typical example. The data were obtained from permeation through a tubular membrane in two opposite directions (Sec. II C 1). The temperature dependencies of two corresponding curves are basically similar, and they give the absolute values of α_{H_2} close to the other; the rather small difference in the absolute value can be attributed to the error in pressure measurements by different gauges. Such observations, first, give the grounds to equally refer the data on α_{H_2} obtained at varying T_S (Figs. 3–5) to the thermostatic conditions. Secondly, this means that α_{H_2} is determined by T_S and not dependent on T_g, at least in the first approximation. These two conclusions were also derived by another way (Sec. II C 1): the values of α_{H_2} evaluated from the steady-state permeation rate at T_g=const=300 K were close to those estimated for $T_g = T_S$, from the permeation lag time τ [see Eqs. (3) and (4) and Fig. 2].

C. Oxygen segregation from bulk onto the surface

The surface analysis by means of AES showed that oxygen was the sole impurity on Nb surface after the treatment described in Sec. II D. Figure 7 shows the temperature dependence of θ_O at different C_O determined from the peak-to-peak intensity ratio of the O *KLL* peak (512.7 eV) to the Nb *MNN* peak (166.5 eV) accounting the relative sensitivity factors of oxygen and Nb (Ref. 28) and the escape depth of Nb *MNN* Auger electrons in Nb evaluated by the method proposed by Seah and Dench.[29]

An evident difference of the presented results from what follows from the Langmuir-McLean equation [Eq. (1)] appears to be most important in the context of the present work. According to Eq. (1), a saturation coverage $\theta_O \approx 1$ is to be achieved by decreasing T_S *for all* C_O. Another picture, however, is actually observed: although θ_O ceases to grow by decreasing T_S, the achieved constant level of θ_O turns out to be noticeably *different for different* C_O.

Niobium is known to form oxide phases of different stoichiometries: NbO to Nb_2O_5. Hence, an important question arises: What is the stoichiometry of the ultimate coverage formed by the surface segregation of solute oxygen? The maximum oxygen coverage at C_O as large as 2 at. % was evaluated as $NbO_{0.9}$ (Fig. 7). Thus, one can conclude that the ultimate segregated coverage was close to NbO. The same inference can be derived from Rieder's data for Nb(110) (Ref. 8) as well as from the recent studies by scanning tunneling microscopy on Nb(110) (Ref. 30) and Nb(100) (Ref. 31). There are grounds to assume that the same *surface oxide* is formed also at lower C_O but with a higher concentration of the holes in coverage. For example, the oxygen coverage at C_O=0.2 at. % is smaller by 23% than at C_O=2 at. % (Fig. 7), and this difference can be attributed to the concentration of the holes in coverage. Notice that, according to our experience, oxygen concentration of 0.1–0.2 at. % is typical of a usual commercially available Nb. Hence, one can presume the layer of segregated oxygen to be typically rather *holey*

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(with a hole concentration many orders of magnitude greater than that expected from the Langmuir-McLean equation at low T_S).

The literature data on the surface segregation of oxygen in the O–Nb system are rather contradictory. Unlike the present data, Hofman *et al.*[32] have reported a complete agreement of the AES data with the Langmuir-McLean equation: the segregated oxygen coverage on polycrystalline Nb did, according to the authors, reach the same value $\theta_O \approx 1$ at T_S decreasing below 670 K in a wide range of C_O, including C_O as low as 1×10^{-3} at. %.

In contrast to that, a number of earlier works indicate that the behavior of oxygen coverage does not obey Eq. (1) at low enough T_S for the O–Nb system. The reduction in low temperature oxygen coverage with decreasing C_O was observed by Farrel *et al.*[7] and Joshi and Strongin.[10] Rieder[8] also reported that the coverage obtained at low T_S was significantly smaller than the monolayer at $C_O = 0.1$ at. %. The present data at selected C_O are compared with the results of these researchers in Fig. 7. The temperature dependencies of data presented are similar, but the absolute values of θ_O differ from each other especially at low C_O. The most probable cause of this disagreement is, in our opinion, the inaccuracy in the determinations of C_O by electrical resistivity[7,10,32] or by microhardness.[8] The present samples are not free from such a sort of inaccuracy either because the samples controllably doped with oxygen were additionally oxidized during the transfer into the AES chamber (Sec. II D). Uncontrolled amount of oxygen introduced by the dissolution of this oxide film was evaluated to be less than 0.1 at. %.

Notice that cessation of θ_O growth with decreasing T_S was observed at rather high T_S (Fig. 7), which by no means can be attributed to the "freezing" of surface segregation of oxygen from Nb bulk: the sample heating after sputter cleaning showed that the segregation of oxygen occurred quickly even at $T_S \sim 700$ K and, according to Rieder,[8] it actually becomes frozen at only $T_S < 520$ K.

IV. DISCUSSION

A general remark about the possible role of the polycrystalline structure of samples seems to be worthwhile in the beginning. We used recrystallized polycrystalline samples with grain size on a millimeter scale and with (100) plane mostly coming out onto the surface. The use of the polycrystalline rather than monocrystal samples does not appear to be of prime importance in the present work. In the first place, it must be on the whole less important for chemical reactions at a metal surface covered by nonmetallic impurities (e.g., because of an impurity induced surface reconstruction which produces a strongly heterogenic surface structure—even on what initially was an ideal monocrystal face[30,31]). Besides, the change in the concentration of solute oxygen had an effect so strong on the surface reaction kinetics (Figs. 3 and 5) that the specificity of a certain face orientation hardly can play a comparable role. As for the grain boundaries, they may be regarded as a sort of heterogeneity elements, and one could expect them to increase the reaction rates. The sticking coefficient of H_2 molecules obtained in the present experiment, however, was extremely small (especially, at high C_O), e.g., orders of magnitude smaller than those found for V(111) (Ref. 4) covered by oxygen and V(100) (Ref. 5) by oxygen and carbon (with uncertain magnitudes of C_O).

A. H_2 sticking to oxygen-covered Nb

1. The concept of coverage holes as active centers of H_2 sticking to metal covered by segregated impurity

The pronounced dependence of α_{H_2} on C_O observed over the whole range of T_S down to room temperature (Figs. 3 and 5) clearly indicates that not the holes in barrier but the holes in coverage act as the reaction centers determining the main reaction path (see Sec. I). There is also an evident correlation between the cessation of reduction in the oxygen-free portion of the surface with decreasing T_S (Fig. 7) and the Arrhenius plot of $\alpha_{H_2}(T_S)$ having the characteristic bends in the range of 800–1300 K depending on C_O (Fig. 3). One can assume that the steep slope of high temperature branches of the $\alpha_{H_2}(T_S)$ plot is connected with not only a barrier against H_2 sticking but also (and mostly) with temperature dependence of the concentration of holes in the oxygen coverage $(1 - \theta_O)$. Vice versa, the transfer to a much weaker $\alpha_{H_2}(T_S)$ dependence at lower T_S (Fig. 3) can be explained by the cessation of $1 - \theta_O$ change with T_S as observed by AES (Fig. 7) and, correspondingly, the slope of low temperature branches of the Arrhenius plot (Fig. 3) is determined by H_2 sticking itself. Taking these observations and ideas into consideration, we attempted to make the phenomenological description of H_2 molecule sticking to the metal covered by a nonmetallic impurity as presented below. The main goal of this simple model is to show clearly the role of surface impurity dynamics in the dissociative sticking.

Suppose that the metal surface is covered by an impurity, with σ being the portion of surface occupied by the sites active in the dissociative sticking of H_2 molecules. If the temperatures of the gas and the surface are different ($T_g \neq T_S$), the probability of sticking to an active site may be determined, in the general case, by both T_g (i.e., translational and internal energies of gas molecules) and T_S (i.e., vibrational energy of surface atoms). The roles of T_g and T_S were found to be nearly equal in the sticking of H_2/D_2 molecules to a clean Cu(111) surface.[13] To reveal the peculiarities of an impurity-covered surface more clearly, we will consider a simpler case when only one of T_g and T_S is playing the prevalent role in the activation of H_2 sticking to active sites (e.g., the latter case actually takes place in the present study, as well as in Refs. 1, 22, 26, and 27). Here, below, we introduce $T_{S,g}$ to designate the temperature playing the dominant role: $T_{S,g} = T_S$ if the dissociative sticking to an active site is governed by surface vibration, while $T_{S,g} = T_g$ if the translational or internal energy of gas molecules controls the sticking. The integral sticking coefficient of H_2 molecules can be represented, in accordance with the general surface reaction kinetics rules,[1] as

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$$\alpha_{H_2} = (0.1-1)\sigma \exp\left(-\frac{2E_b}{RT_{S,g}}\right), \quad (6)$$

where $2E_b$ is the activation energy of sticking at the active sites and (0.1–1) is the typical range of kinetic coefficient (conditioned, e.g., by the preferential sticking of molecules having certain orientations[14]).

If the active portion of surface, σ, is determined by the sites free from impurity (oxygen in the present case), one should not consider σ as a coefficient independent of T_S: the number of oxygen-free sites, $1-\theta_O$, depends on T_S (and C_O) according to Eq. (1) at high enough T_S. One can suppose that $\sigma=(1-\theta_O)^n$, where n depends on the number of adjacent oxygen-free sites forming one active center (e.g., $n=2$, if double oxygen vacancies make up active centers for the dissociative sticking of diatomic molecules). With taking, for a typical case, $\theta_O/(1-\theta_O)\approx 1/(1-\theta_O)$ in Eq. (1) and inserting the result to Eq. (6), one can get

$$\alpha_{H_2} = (0.1-1)\frac{1}{k_O^n C_O^n}\exp\left(-\frac{n|\Delta H_{seg}|}{RT_S}-\frac{2E_b}{RT_{S,g}}\right) \quad \text{at } T_S > T_S^*(C_O), \quad (7)$$

where T_S^* is the lowest temperature at which Eq. (1) is valid.

In reality, the reduction in the portion of oxygen-free surface, $1-\theta_O$, with decreasing T_S [Eq. (1)] will stop at a certain temperature T_S^* (T_S^* may depend on C_O). This is inevitable, at least due to the decreasing impurity diffusivity. In the present case, that happened at T_S^* much higher than that expected due to the limitation by diffusivity, as described in Sec. III C. In any case, when σ ceases to change with temperature at $T_S<T_S^*$, the dependence of α_{H_2} on C_O still holds [cf. Eq. (7)],

$$\alpha_{H_2} = (0.1-1)q(C_O)\frac{1}{k_O^n C_O^n}\exp\left(-\frac{2E_b}{RT_{S,g}}\right) \quad \text{at } T_S < T_S^*(C_O), \quad (8)$$

where $q=\exp(-n|\Delta H_{seg}|/RT_S^*(C_O))$.

Notice that Eqs. (6)–(8) are all valid under the thermostatic conditions ($T_g=T_S=T$) as well.

According to Eqs. (7) and (8), the Arrhenius plot of $\alpha_{H_2}(T_S)$ [or $\alpha_{H_2}(T)$ under thermostatic conditions] is expected to bend at $T_S=T_S^*$, with the "actual barrier" ($2E_b$) determining $\alpha_{H_2}(T_S)$ at $T_S<T_S^*$ and with a larger apparent activation energy ($n|\Delta H_{seg}|+2E_b$) at $T_S>T_S^*$. As described above, $2E_b$ is the activation energy for sticking to the holes in coverage and not to the regular surface sites covered by oxygen. However, the presence of a definite concentration $1-\theta_O$ of such holes, determined by T_S and C_O, is a basic property of a metal-impurity system. Therefore, if the holes in coverage do actually serve the active centers, $2E_b$ is the activation energy ("barrier") that is expected to determine the kinetics of hydrogen interaction with a metal-impurity system. The most important implication of Eqs. (7) and (8) (as well as of Figs. 3 and 5) is that α_{H_2} can depend on C_O at *all* temperatures, including the range $T_S<T_S^*$ where the concentration of coverage holes does not change with T_S.

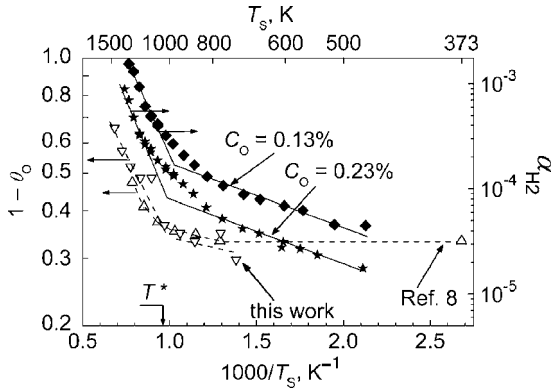


FIG. 8. Arrhenius plot of the portion of surface free from oxygen $1-\theta_O$ (the left axis, open symbols) and of α_{H_2} (the right axis, solid symbols). The values of $1-\theta_O$ was calculated from the AES analysis data presented in Fig. 7.

2. Correlation between α_{H_2} and oxygen coverage

It should be noted in the first place that oxygen segregated onto the surface forms no more than a monolayer coverage even at highest C_O and lowest T_S, as observed by Joshi and Strongin,[10] who studied the segregation of oxygen from Nb bulk onto the surface with C_O varying up to $C_O=2.3$ at. % and T_S down to 300 K. Another argument to corroborate this notion is that the sticking coefficient of H atoms was comparable to 1, not depending on C_O and T_S (Fig. 5), which is characteristic of a coverage not exceeding a monolayer.[33]

The temperature dependencies of α_{H_2} (Fig. 3) and of the fraction of oxygen-free surface sites, $1-\theta_O$, evaluated from the results of AES analyses (Fig. 7) are presented together in Fig. 8 to show their correlation at similar values of C_O (0.1–0.2 at. %). The plots for α_{H_2} show the bend at similar temperatures (around 1000 K) to those for $1-\theta_O$. Notice that both the temperature T_S^* at which the Arrhenius plot of $\alpha_{H_2}(T_S)$ exhibit the bend and the temperature where oxygen coverage θ_O ceases to increase with decreasing T_S rise with increasing C_O. Such a clear correlation between the temperature dependencies of α_{H_2} and the fraction of oxygen-free surface sites serves one more evidence in favor of the holes in coverage concept described above.

It should be noted that the increase of T_S^* with C_O means that the cessation of θ_O growth at $T_S<T_S^*$ is not due to the slow diffusion of oxygen in Nb; oxygen can be gathered from a smaller depth at higher C_O. A simple evaluation based on the data on the diffusion coefficient of oxygen in Nb, $D_O=1.5\times10^{-6}\exp(-117\ \text{kJ/mol}/RT_S)\ \text{m}^2/\text{s}$ (Ref. 19), also indicated that the diffusion of oxygen was fast enough at T_S^* observed in the present study.

The increase of T_S^* with C_O leads to the increase of $q[T_S^*(C_O)]$ in Eq. (8); as a result, the dependence of α_{H_2} on C_O is expected to be stronger at higher temperatures [$\alpha_{H_2}\propto 1/C_O^n$ at $T_S>T_S^*$, Eq. (7)] than at lower temperatures [$\alpha_{H_2}\propto q(C_O)/C_O^n$ at $T_S<T_S^*$, Eq. (8)]. The observed dependence of α_{H_2} on C_O does actually demonstrate such a behavior (Fig. 5). At $T_S>T_S^*$, $\alpha_{H_2}(C_O)\propto 1/C_O^n$, whence it follows that the exponent n in Eqs. (7) and (8) equals 2. One can interpret $n=2$ as the sticking of diatomic molecules to active centers

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formed by double oxygen vacancies. At $T_S < T_S^*$, the dependence of α_{H_2} on C_O could be approximated by $\alpha_{H_2}(C_O) \propto 1/C_O^{1.5}$ (Fig. 5).

According to the phase diagram of the Nb–O system,[19] oxygen in the bulk only can be in the state of solid solution at $T_S > T_S^*$ in the whole investigated range of C_O. Niobium oxide (NbO) should be still formed at lower T_S, e.g., at $T_S < 690$ K with $C_O = 0.2$ at. %, but that had no noticeable effect on temperature dependencies of both the concentration of the holes in coverage $1-\theta_O$ (Fig. 8) and α_{H_2} (Figs. 3 and 8). The similarity in the $\alpha_{H_2}(C_O)$ curves at different T_S (Fig. 5) provides another evidence that the role of oxygen solubility limit is insignificant; the data at 970 and 1120 K, where oxygen only can be in the state of solid solution, are basically similar to those at 770 and 570 K, where NbO should precipitate at high C_O (e.g., with $C_O \geqslant 0.3$ at. % at 770 K).[19] Such a behavior of $\alpha_{H_2}(C_O)$ can be explained by that the concentration of the holes in coverage at $T_S < T_S^*$ corresponds to the equilibrium at $T_S \approx T_S^*$, i.e., at a relatively high temperature where bulk oxygen is only in the solid solution phase.

3. Activation energy for dissociative sticking

First, note that the Arrhenius plot shown in Fig. 3 gives another evidence (see also Sec. II C 3) in favor of that the surface of oxygen-covered Nb was not saturated by hydrogen atoms (hydrogen coverage $\theta_H \ll 1$) even at the lowest T_S (300 K). In fact, no significant change was observed in the slope of the Arrhenius plot from T_S^* down to 300 K. Therefore, taking into account that the surface was certainly almost free from hydrogen at the temperatures as high as T_S^*, one can surmise that saturation did not occur at lower T_S as well.

Taking into account the observation that α_{H_2} depends on T_S and not on T_g (Sec. III B), one can derive an apparent activation energy for H_2 sticking at $T_S < T_S^*$ as $2E_b = 10 \pm 2$ kJ/mol from the $\alpha_{H_2}(C_O, T_S)$ plot in Fig. 3 with using Eq. (8). The data obtained at room temperature were not included in this evaluation because they were less accurate as described above. This activation energy relates to the sticking at the holes in coverage (double oxygen vacancies as discussed above). On the other hand, the activation energy for sticking to the *regular surface sites covered by oxygen*, $2E_b^{reg}$, appears to be very high because the values of α_{H_2} were small even at high T_S and T_g at large C_O, e.g., $\alpha_{H_2} = 2 \times 10^{-5}$ at $T_S = 1300$ K and $T_g = 300$ K at the maximum C_O (Fig. 3), with similar values of α_{H_2} under thermostatic conditions $T_g \cong T_S = T = 1350$ K (Sec. III B). Since the observed sticking is determined by holes in the coverage, the sticking coefficient to the regular surface sites covered by oxygen, $\alpha_{H_2}^{reg}$, must be much smaller than 2×10^{-5} at $T = 1350$ K. On accepting $\alpha_{H_2}^{reg} = \exp(-2E_b^{reg}/RT)$, we can roughly estimate the activation energy for sticking to the *oxygen-covered regular surface sites* as $2E_b^{reg} > 100$ kJ/mol. This shows that an adlayer of oxygen on the transition metal surfaces (Nb in the present case) is able to generate a barrier at least twice as large as the barrier at a clean surface of the group XI metals[13,14,34] whose low reactivity is due to the filled *d* band.

In the present study, there was no indication of the presence of "holes in barrier" at the regular surface sites covered by oxygen. A high barrier, but with "holes" responsible for hydrogen sticking/desorption, was found on V(111) covered by oxygen and sulfur by Eibl and Winkler.[4] These authors, however, did not consider the "holes in coverage" as a possible alternative to interpret their results.

The apparent activation energy at $T_S > T_S^*$ was >110 kJ/mol (Fig. 3). Assuming $n=2$ and $2E_b = 10 \pm 2$ kJ/mol, one can get $|\Delta H_{seg}| > 50$ kJ/mol by Eq. (7). This value does not contradict to those found by Farrel *et al.*[7] (71 kJ/mol), Joshi and Strongin[10] (46–71 kJ/mol), Hofmann *et al.*[32] (50–63 kJ/mol), and Rieder[8](63 kJ/mol).

4. Preexponential factor in the sticking coefficient

According to Eq. (6) where the active portion of the surface, σ, is simply assumed to be independent of $T_{S,g}$, the preexponential factor of α_{H_2}, $\alpha_{H_2}^*$, should be always smaller than 1. This is usually observed:[1,3,21,22,26,27,35,36] $\alpha_{H_2}^*$ was varying over a range of 10^{-4}–10^{-1} for different metals covered by different impurities. If, however, the active centers of dissociative sticking are the holes in coverage whose concentration changes with T_S at $T_S > T_S^*$, then α_{H_2} is described by Eq. (7), and hence a dependence of $\alpha_{H_2}^*$ on C_O appears,

$$\alpha_{H_2}^* = (0.1-1)\frac{1}{k_O^2 C_O^2} \quad \text{at } T_S > T_S^*. \tag{9}$$

With counting C_O in an atomic fraction, we should get the factor k_O in Eq. (1) in the order of 1.[37] Hence, one can expect $\alpha_{H_2}^* \gg 1$ at $C_O \ll 1$, but not $\alpha_{H_2}^* \leqslant 1$ as that is usually observed. The values of $\alpha_{H_2}^*$ at $T_S > T_S^*$ evaluated from the data in Fig. 3 turned out to be in the range of 0.15–50 depending on C_O; the values larger than unity were obtained. Still more accurate measurements of the apparent activation energy are required to make more reliable conclusions about the $\alpha_{H_2}^*$ behavior at $T_S > T_S^*$.

At $T_S < T_S^*$, the values of $\alpha_{H_2}^*$ observed were much smaller than 1 and decreased with increasing C_O [in accordance with Eq. (8)] from 10^{-2} to 10^{-4}. This is another evidence that relatively scarce active centers are associated with holes in the coverage.

5. The role of T_g and T_S

The observed decisive role of T_S in H_2 sticking (Sec. III B) does not seem to be a unique property of oxygen-covered Nb: other transition metals with the surface covered by nonmetallic monolayers exhibit the same pattern of behavior. For example, this was observed for H_2/D_2 sticking to polycrystalline Pd covered by sulfur and carbon,[1] as well as to polycrystalline Fe,[26] Pd,[22] Ni,[27] and Nb (Ref. 1) covered by residual impurities (most probably, by sulfur, oxygen, and carbon).

If the activated sticking is determined by T_S, the desorption flux will mostly consist of thermal molecules having a

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temperature T_S (in accordance to the detailed balance principle). The above has been confirmed for the oxygen-covered Nb (Sec. III B) by the agreement of α_{H_2} found from the direct process (steady-state permeation) with varying T_S at T_g=const=300 K with that evaluated from the desorption rate constant k_r, with applying the detailed balance principle. Similar agreement was observed also for other metals covered by a monolayer of nonmetallic impurity.[1,22,26,27] It is to be noted, however, that D_2 flux desorbing from V(111) covered by oxygen and sulfur at T_S=950 K (Ref. 4) was observed to consist of more than a *half* of hyperthermal D_2 molecules having the average translational energy exceeding $2kT_S$ by a factor of 3–4 [although the average translational energy exceeded $2kT_S$ only slightly, by a factor of 1.25–1.5, at the desorption from oxygen-covered V(100)].[38]

The difference between the sticking of H_2 to oxygen-covered Nb (as well as to other impurity-covered metals mentioned above) and the activated sticking to a clean surface of Cu or Ag should be emphasized. Approximately equal influence of T_g and T_S on H_2 sticking probability was reported for Cu(111) and Ag(111),[13,34] although escaping from the surface were virtually *only hyperthermal* molecules. The explanation is that the surface vibration (T_S) creates surface atom configurations with a lower repulsive barrier to be overcome by an impinging molecule due to its translational energy (T_g).[13]

The decisive role of T_S in H_2 sticking to oxygen-covered Nb (as well as to other impurity-covered metals) may be explained in the following way. The holes in the coverage (supposedly, double oxygen vacancies) serve as the dissociation sites responsible for the reaction kinetics.[2,17,18] However, H_2 molecules stick at these holes in coverage, only if the holes have a specific configuration with a repulsive barrier equal to zero (at least at a fraction of area occupied by the hole in coverage) and that is why α_{H_2} did not depend on T_g. Such configurations of the holes in coverage with zero barrier emerge due to the thermal activation (the surface vibration at temperature T_S) with an activation energy $2E_b \approx 10$ kJ/mol observed at $T_S < T_S^*$ (Sec. IV A 3). Thus, one can say that the molecules stick to the "holes in holes," i.e., to holes in the barrier (according to the definition of Ref. 39) distributed over the holes in coverage.

An alternative explanation for the negligible role of T_g is that H_2 molecules first stick with no activation energy into a predissociation state to be thermalized there. Then, they dissociate at coverage holes with an activation energy $2E_b \approx 10$ kJ/mol. The coefficient α'_{H_2} of sticking into a such precursor state should be small enough to be compatible with the energy accommodation coefficient β which usually is substantially smaller than 1 [β=0.37 for V(111) covered by oxygen and sulfur],[4] i.e., there should be $\alpha_{H_2} \ll \alpha'_{H_2} \ll \beta$.

It should also be noted that the negligible effect of T_g on α_{H_2} does not mean that the translational or the vibrational energy has no promotional effect whatsoever on the sticking. It only means that gas energy facilitates sticking not effectively enough (if it ever does) to make a noticeable contribution to sticking in the case of the equilibrium energy distribution (i.e., the concentration of energetic molecules is simply too small even at the maximum T_g investigated). For instance, if one refers the data in Fig. 3 to the thermostatic conditions (see Sec. III B), α_{H_2} at $T_g=T_S=1300$ K is as small as $\approx 2\times10^{-5}$ (for C_O=0.2 at. %), while the equilibrium concentration of vibrationally excited $H_2(\nu=1)$ molecules is $\sim$0.01 at T_g=1300 K. Hence, one gets a rather low sticking coefficient of $H_2(\nu=1)$ molecules to oxygen-covered Nb: $\alpha_{H_2}(\nu=1) < 2\times10^{-3}$ [whereas the sticking coefficient of H atoms α_H was a few tenths (!) as shown in Fig. 5]. This still does not exclude the possibility of $\alpha_{H_2}(\nu=1) \gg \alpha_{H_2}(\nu=0)$, e.g., $\alpha_{H_2}(\nu=1) > 10^{-4}$ [correspondingly, the desorption flux, while remaining mainly thermal, will be greatly overpopulated with $H_2(\nu=1)$ in this case]. Notice that the same is valid for atoms: although their sticking probability is as high as a few tenths, they make no input into the integral sticking at not very high T.

B. Sticking of H atoms at oxygen-covered Nb

There is an essential difference between the effects of oxygen coverage upon the sticking probabilities of atoms, α_H, and of molecules, α_{H_2} (Fig. 5): α_{H_2} decreased by orders of magnitude as compared to the clean surface with increasing C_O and decreasing T_S, while α_H stayed in a few tenths (i.e., comparable to 1), depending neither on C_O nor on T_S. The large values of α_H and the insensitivity of α_H to C_O indicate that, unlike H_2 molecules, H atoms are mainly absorbed via the regular surface sites covered by oxygen and not through the holes in coverage.

Sticking of H atoms with a probability of a few tenths is quite typical of metals covered by nonmetallic monolayers: it was observed on polycrystalline Pd covered by sulfur and carbon,[1] for V(100) covered by oxygen and carbon,[5] V(111) covered by sulfur and oxygen,[4] as well as for polycrystalline Fe (Ref. 26) and Ni (Refs. 1 and 27) covered by residual impurities (supposedly by oxygen, carbon, and sulfur). Typical of α_H also is its independence of T_S, at least at elevated temperatures ($T_S > 400-500$ K).[1,17,18,26,27]

The probability of the absorption of atoms was much larger than that of molecules but still distinctly smaller than unity. It was found[1,22,40] that a major fraction of the atoms that are not absorbed escapes from the surface with retaining atomic state, at least at elevated temperatures. The fact that the majority of impinging atoms scatters with no absorption (or recombination) may be explained by the chemical and structural microheterogeneity of the surface modified by chemisorbed impurity. For instance, a major part of the surface unit cell may be arranged in such a way that the atoms impinging upon it cannot be chemically bonded (e.g., like He atoms) and are almost elastically scattered at the investigated temperatures ($T_S > 300$ K). One would expect the energy accommodation coefficient β for the scattered atoms to be in a few tenths in this case. Another possibility may be that the atoms impinging upon the "passive" part of unit cell are bonded, but rather weakly, and are ultimately desorbed in the form of atoms at $T_S > 300$ K because the desorption probability is still greater than the probability of dissolution into the lattice (i.e., there is a high enough barrier between the states of adsorption and solution). The desorbed atoms pos-

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sess the temperature of the surface in this case, i.e., β is expected to be close to 1. Sticking of H atoms into a weakly bonded surface state was observed in the case of V(100) covered by residual impurities (a monolayer of oxygen and carbon):[5] at T_S=90 K, atoms were sticking with a probability greater than 0.6 and then released at heating to $T_S \approx 150$ K (though it is not clear whether as H or H_2).

C. Sticking to the clean surface

A remarkable α_{H_2} invariance was observed in the range of T_S as wide as 300–1800 K. A similar lack of dependence of α_{H_2} on T_S over a very wide range has been also observed for W: $\alpha_{H_2} \approx 0.25$ was found for W(111) in the range of T_S 78–425 K,[41] and the same value was obtained for $T_S > 1800$ K from the dissociation kinetics of H_2 molecules at the surface of polycrystalline W.[25] One can suppose that the lack of dependence of α_{H_2} on T_S is universal for the case when there is a reaction path with no activation barrier[14] (i.e., for most of crystal faces of many transition metals). In contrast to that, a strong exponential increase of α_{H_2} and α_{D_2} with T_S was found for Cu(111) (Ref. 13) and Ag(111) (Ref. 34) because the whole surface is blocked by barriers.

Although α_{H_2} for Nb is found to be large ($\sim$0.25), it is still substantially smaller than 1. This is supposed to be so because only a part of the unit cell is free from barrier (the fraction of this part depends, e.g., on the incident molecule orientation).[14] In principle, the thermal vibration of surface atoms could enhance the barrier-free part. One can conclude from the lack of dependence of α_{H_2} on T_S that the contribution of such a mechanism is negligible when a significant part of unit cell is barrier-free and, consequently, α_{H_2} is large even at a static surface. In contrast to that, formation of surface atom configurations with a lower barrier due to surface motion was found to be the cause of strong exponential growth of $\alpha_{H_2}(\alpha_{D2})$ with T_S in the case of activated sticking to a clean surface [e.g., to Cu(111), Ag(111)].[13]

V. CONCLUSIONS

Hydrogen molecules were found to stick to, and be dissolved in, oxygen-covered Nb with a probability α_{H_2} strongly depending on oxygen concentration C_O in the bulk at all investigated T_S (300–1620 K). Therefore, the main channel of H_2 sticking to oxygen-covered Nb is the dissociation at the holes fundamentally inherent to the segregated oxygen coverage, while the regular sites occupied by oxygen atoms have a barrier insuperable for thermal H_2 molecules (>100 kJ/mol). There was no evidence of holes in this barrier (i.e., of strongly corrugated potential energy surface). The active centers responsible for H_2 sticking are most likely associated with double oxygen vacancies in the oxygen monolayer. The surface vibration (T_S), and not the translational or internal energy of incident molecules (T_g), played the dominant role in the activation of H_2 molecule dissociative sticking to oxygen-covered Nb (at least up to 1620 K under the thermostatic conditions). These observations indicate that the concentration of the bulk impurity are to be taken into account to characterize the reactivity of metals, even at low T_S (e.g., 300 K) and even in the cases where the surface composition is well defined. The fact that α_{H_2} is determined by the bulk impurity concentration provides us with a means to control the barrier of dissociative sticking (and, correspondingly, the barrier of recombinative release) of H_2 molecules.

Hydrogen atoms were found to stick to, and be dissolved in, oxygen-covered Nb with a high probability (in a few tenths), depending neither on C_O nor on T_S. Thus, in contrast to H_2 molecules, H atoms stick to oxygen-covered Nb through the regular surface sites covered by oxygen (and not through the holes in coverage).

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